Ravleen Bajaj

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Education

Simon Fraser University *Expected Graduation: August 2027*
MS in Statistics *CGPA: 3.76/4.33 (≈ 3.74/4.0)*

- **Selected Coursework:** Statistical Theory I (A), Statistical Theory II (A; measure-theoretic probability), Statistical Research & Inverse Problems (A), Statistical Consulting I (A+), Linear Models and Applications (A), Applications of Statistical Computing (A-), Statistical Consulting II (A).

University of Delhi *Jul 2021 – Jul 2024*
BS (Honours) in Statistics *CGPA: 8.203/10, First Division*

- **Selected Upper-Level Coursework:** Sampling Distributions (A), Time Series Analysis (A), Multivariate Analysis and Nonparametric Methods (A+), Linear Models (A), Design of Experiments (A), Statistical Computing Using C/C++ Programming (A).

Experience

Incoming Applied Machine Learning Engineer Intern *Toronto, ON*
Shopify *Jan 2027 – Apr 2027*

Graduate Research Assistant (Master's Thesis) *Burnaby, BC*
Supervised by Dr. David Stenning, Simon Fraser University *Sep 2024 - Aug 2027 (Expected)*

- Bayesian computer model emulation and calibration using (deep) Gaussian Processes (GPs) to infer a star's physical properties from its photometry.
- We examine tradeoffs between modular and non-modular approaches when doing calibration (i.e., inference).
- To efficiently incorporate the high-dimensional and non-stationary simulator response surface, we fit a DeepGP emulator on simulated stellar evolution data. We use Vecchia approximation for faster computation.
- We adopt a multi-stage, modular approach to emulation & calibration, for pragmatic inference while retaining uncertainty quantification. We validate MCMC convergence using trace plots, ESS & R-hat statistic.

Research Assistant *Burnaby, BC*
SFU Statistical Consulting Service *Jan 2026 – Present*

- Statistical analysis for a program evaluation of professional avalanche training, using logistic regression, chi-square tests and Tukey-Kramer post hoc comparisons to assess associations between proficiency classes and participant, course, instructor, and hazard factors. Acknowledged in the resulting ISSW 2026 *paper*.

Graduate Teaching Assistant *Burnaby, BC*
Simon Fraser University *Sep 2024 – Present*

- **STAT 350:** Weekly **lecture-style** *tutorials*. TA in the STAT workshop, a drop-in facility for students.
- Resolving queries, helping students build concepts along with discussing issues with marking and understanding exam solutions. Marking assignments, quizzes, exams and invigilating tests.

Talks and Posters

Canadian Statistics Student Conference (CSSC) *2026*
Emulation & Bayesian Calibration of Stellar Evolution Models with Gaussian Processes

Canadian Celebration of Women in Computing West (CAN-CWiC West) *2025*
Introduction to Gaussian Processes for Emulation and Calibration

Awards and Scholarships

2025 Graduate Fellowship ($7,000 CAD)
2026 Graduate Fellowship ($3,500 CAD)
2026 CSSC Travel Award ($150 CAD)
2026 Graduate Fellowship ($7,000 CAD)
2026 Travel and Research Award ($600 CAD)

Organizing & Scientific Committee

President: Department of Statistics Graduate Caucus *2025-2026*
Co-Organizer: SFU-UBC Joint Statistics Seminar *Fall 2025*
Co-Organizer: SFU-UBC Joint Statistics Seminar *Spring 2026*

Projects

Annealed Sequential Monte Carlo (SMC) for Breast Cancer Detection *2025*

- Implemented an annealed SMC sampler with an adaptive, ESS-based temperature schedule for Bayesian logistic regression.
- Benchmarked against MLE logistic regression on a held-out test set: matched MLE accuracy (97.1%) with credible intervals that don't rely on asymptotic normality. Diagnosed the sampler with ESS, acceptance rate, and the annealing schedule.

Reduction of Small-Sample Bias of GLM Parameter Estimates *2025*

- We implemented and compared three bias-reduction methods for GLM maximum likelihood estimates: asymptotic bias correction, Firth's penalized likelihood method, and Bayesian-inspired log-F priors.
- We conducted simulation studies in R to evaluate estimator bias, efficiency, and coverage probabilities across sample sizes.

Additional Experience

Analyst Intern *New Delhi, IN*
Perfora *Aug 2023 – Oct 2023*

- Developed a web scraping solution to monitor product placements on the website and recommend dynamic placement strategies to improve product and product-combination sales, while tracking overall sales.

Data Science Intern *New Delhi, IN*
DMI Finance *June 2022*

- Developed end-to-end machine learning model reports and documented model inferences for technical and non-technical stakeholders, covering models including XGBoost and logistic regression.

Volunteering

Management Lead *New Delhi, IN*
Google Developer Student Club *2021-2022*

- Coordinated events, and hackathons; delivered "Data Science 101," an introductory talk on Python.

Changemaker Program Volunteer, Analytics Volunteer *New Delhi, IN*
Pratisandhi *Sep 2021-June 2022*

- Cleaned data, performed EDA and conducted analysis on surveys conducted internally and on-ground.

Eco Commander *New Delhi, IN*
Paryavaran Sanrakshan Gatividhi (Govt. of India Initiative) *Oct 2021*

- Achieved "platinum" certification where I educated 100+ individuals about sustainable homes and lifestyle.

Skills / Technologies

Programming: Python, R, Julia, LEAN, SQL, C/C++, bash
Frameworks: DeepGP, PyMC, NumPyro, JAX, TensorFlow
Tools: Git, Slurm, LaTeX, MySQL